**WEEK-1**

Name: Venkata Sai Krishna

Roll No. : 2520030027

SEC: 04

1. Create a class Student with student ID, name and age. Create three student objects and display their details.

```
class Student {
   int studentId;
   String name;
   int age;

   void display() {
      System.out.println("Student ID: " + studentId);
      System.out.println("Name: " + name);
      System.out.println("Age: " + age);
      System.out.println();
   }

   public static void main(String[] args) {
      Student s1 = new Student();
      s1.studentId = 101;
      s1.name = "VSK";
      s1.age = 18;
```

```
        Student s2 = new Student();
        s2.studentId = 102;
        s2.name = "Rahul";
        s2.age = 21;

        Student s3 = new Student();
        s3.studentId = 103;
        s3.name = "Vishwa";
        s3.age = 19;

        s1.display();
        s2.display();
        s3.display();
    }
}
```

2. Create an Employee class with employee ID, name and salary. Initialize the values using a constructor and display employee details.

```
class Employee {
    int employeeId;
    String name;
```

```
    double salary;

    Employee(int id, String n, double s) {
        employeeId = id;
        name = n;
        salary = s;
    }

    void display() {
        System.out.println("Employee ID: " + employeeId);
        System.out.println("Name: " + name);
        System.out.println("Salary: " + salary);
    }

    public static void main(String[] args) {
        Employee e1 = new Employee(101, "VSK", 30000);
        Employee e2 = new Employee(102, "Rahul", 40000);

        e1.display();
        e2.display();
    }
}
```

3. Create a BankAccount class with account number and balance. Implement deposit and withdrawal methods.

```
class BankAccount {
    int accountNumber;
    double balance;

    BankAccount(int accNo, double bal) {
        accountNumber = accNo;
        balance = bal;
    }

    void deposit(double amount) {
        balance = balance + amount;
        System.out.println("Deposited: " + amount);
    }

    void withdraw(double amount) {
        if (amount <= balance) {
            balance = balance - amount;
            System.out.println("Withdrawn: " + amount);
        } else {
            System.out.println("Insufficient Balance");
        }
    }

    void display() {
```

```
        System.out.println("Account Number: " + accountNumber);
        System.out.println("Balance: " + balance);
    }

    public static void main(String[] args) {
        BankAccount b1 = new BankAccount(12345, 5000);

        b1.display();
        b1.deposit(2000);
        b1.withdraw(1000);
        b1.display();
    }
}
```

4. Create a Rectangle class with length and width. Calculate and display its area and perimeter.

```
class Rectangle {
    double length;
    double width;

    Rectangle(double l, double w) {
        length = l;
        width = w;
```

```
    }

    void calculate() {
        double area = length * width;
        double perimeter = 2 * (length + width);

        System.out.println("Length: " + length);
        System.out.println("Width: " + width);
        System.out.println("Area: " + area);
        System.out.println("Perimeter: " + perimeter);
    }

    public static void main(String[] args) {
        Rectangle r1 = new Rectangle(10, 5);
        r1.calculate();
    }
}
```

5. Create a Book class with ISBN, title and price. Initialize the values using a parameterized constructor.

```
class Book {
    String ISBN;
    String title;
```

```
    double price;

    Book(String i, String t, double p) {
        ISBN = i;
        title = t;
        price = p;
    }

    void display() {
        System.out.println("ISBN: " + ISBN);
        System.out.println("Title: " + title);
        System.out.println("Price: " + price);
    }

    public static void main(String[] args) {
        Book b1 = new Book("9781234567890", "Java Programming", 450);
        Book b2 = new Book("9789876543210", "Python Programming", 500);

        b1.display();
        System.out.println();
        b2.display();
    }
}
```